INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
(School of University of Engineering and Management, Kolkata)

End Semester Examination, November - December, 2025

Degree: **B.Tech** Semester: **First**

Stream: **CSE(AI & ML)/ CSE (Data Science)/ CSE(IOT)/ Robotics & AI/ CSBS/ CSE/ CSIT/ CST/ ECE/ EEE/ EE/ IT/ ME**

Paper Code: **ESCEE101**

Paper Name: **Basic Electrical Engineering**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. Define power factor as applied to ac circuits. What do you mean by active power, reactive power and apparent power? **2**

2. Explain with the help of diagrams what you understand by in phase, lagging and leading as applied to simusoidal quantities. **2**

3. Show that power consumed in a purely inductive circuit and purely capacitive circuit is zero when sinusoidal voltage is applied across it. **2**

4. Define quality factor in a series *R-L-C* circuit. Determine the half power frequencies in terms of quality factor and the resonant frequency for series *R-L-C* circuit. **2**

5. Give the applications of dc choppers. **2**

6. What is a step-up chopper? **2**

7. Enlist a few industrial applications of inverters. **2**

8. What is pulse-width modulation? **2**

9. Why conductor is covered with insulating material? **2**

10. PVC cables are most widely used in internal wiring. Why? **2**

11. What is meant by a switch? **2**

12. What kind of protection is offered by a fuse? **2**

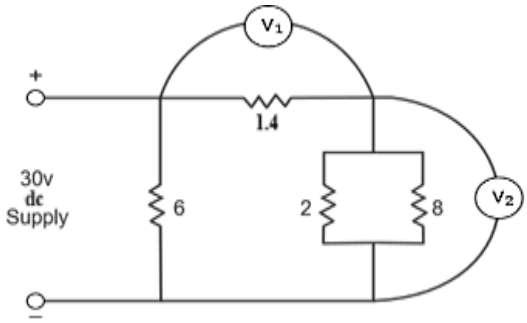
13. What is MCB and what are its application? **2**

14. What is earthing? **2**

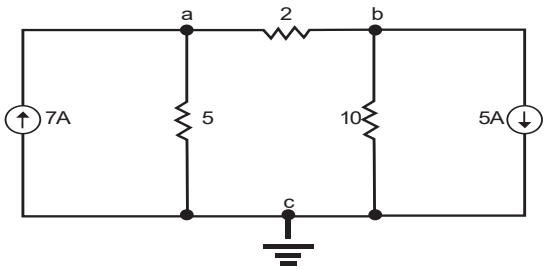
15. Lead acid battery can charged whereas primary cell cannot be charged. Why? **2**

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

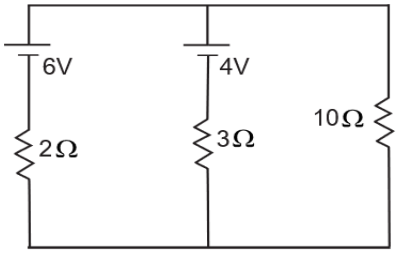
16. Neglecting the current taken by voltmeters V_1 and V_2 in the figure calculate (a) total current taken from the supply (b) reading on voltmeter V_1 and (c) reading on voltmeter V_2. **5**



17. Use nodal analysis to find various branch currents in the circuit of the figure. [Hint: Check by source conversion.][$I_{ac} = 2A$; $I_{ab} = 5A$; $I_{bc} = 0$] **5**



18. Apply the principle of superposition to the network shown in the figure to find out the current in the 10Ω resistance. **5**



19. State and prove maximum power transfer theorem. **5**

20. An ac series circuit consisting of a pure resistance of 25 Ω , inductance of 0.15 H and capacitance of 80 μF is supplied from a 230 V, 50 Hz ac source. **5**

(a) Find the impedance of the circuit, the current, the power drawn by the circuit and the power factor.

(b) Draw the phasor diagram.

21. A resistance of 12Ω and inductance of 0.15 H and a capacitance of 130 μF are connected in series across a 100 V, 50 Hz supply. Determine the impedance, current and power factor of the circuit. **5**

22. Derive an expression for the emf induced in a transformer winding. **5**

23. A 1100/230 V, 150kVA single- phase transformer has a core loss of 1.4 kW and a full load copper loss of 1.6 kW. Determine (i) the kVA load for maximum efficiency and (ii) the maximum efficiency at unity power factor load. **5**

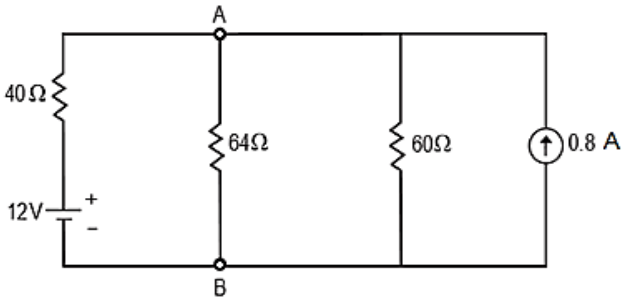
24. A d.c. motor connected to a 460-V supply takes an armature current of 120 A on full load. If the armature circuit has a resistance of 0.25 **Ω,** calculate the value of the back e.m.f. at this load. **5**

Part - C

Attempt any 5 questions out of 8 questions

Each question carries 10 Marks (10 × 5)

25. Using Norton's theorem find the current through 64 **Ω** resistance in the circuit shown in the figure **10**



26. The number of turns on the primary and secondary windings of a single-phase transformer are 350 and 38 respectively. If the primary winding is connected to a 2.2 kV, 50-Hz supply. Determine- **10**

(a) The secondary voltage on no-load.

(b) The primary current when the secondary current is 200 A at 0.8 p.f. lagging, if the no-load current is 5 A at 0.2 p.f. lagging.

(c) The power factor of the primary current.

27. **(a)** A DC-shunt machine has armature and field resistance of 0.025Ω and 80 Ω respectively. When connected to constant 400-V bus-bars and driven as a generator at 450 r.p.m, it delivers 120kW. Calculate its speed when running as a motor and absorbing 120kW from the same bus-bars. **10**

(b) Deduce the direction of rotation of machine when it is working as a motor assuming a clockwise rotation as a generator.

28. On no-load, a shunt motor takes 5 A at 250 V, the resistance of the field and armature circuits are 250 Ω and 0.1 Ω respectively. Calculate the output power and efficiency of the motor when the total supply current is 81 A at the same supply voltage. State any assumptions made. **10**

29. An 8-pole lap-wound d.c. generator armature has 960 conductors, a flux of 40 mWb and a speed of 400 r.p.m. Calculate the e.m.f. generated on open circuit. If the same armature is wave wound, at what speed must it be driven to generate 400 V? **10**

30. A 4-pole, 3-phase induction motor operates from a supply whose frequency is 50 Hz. Calculate **10**

(a) The speed at which the magnetic field of the stator is rotating;
(b) The speed of the rotor when the slip is 0.04;
(c) The frequency of the rotor current when the slip is 0.03;
(d) The frequency of the rotor current at standstill.

31. What is sinusoidal pulse width modulation? How is it obtained? Explain with the help of neat diagram. **10**

32. A step-up chopper has an ac input voltage of 100 V and an output voltage of 250 V. The period of blocking in each cycle is 0.6 ms. Find period of conduction and chopping frequency. **10**

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